An Introduction to Bayesian Methods in Neuropsychology for Psychometrists

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Table of Contents

1 Introduction

1.1 About Me

- **Education**
  - B.S. in Psychology (Neuroscience Concentration) from Calvin University
  - M.S. in Quantitative Psychology from Ball State University
- **Current Positions**
  - Lead Psychometrist at Trinity Health West Michigan Medical Group - Neuropsychology Department
  - Adjunct Professor of Psychology at Grand Valley State University
  - Adjunct Instructor of Educational Psychology at Ball State University

1.2 Disclosures & Disclaimers

- I have no financial disclosures or conflicts-of-interests related to this presentation
- This presentation has not been drafted, ideated on, or edited with use of AI or LLMs. Any mistakes made are human error.
- CE credits for viewing this presentation will be provided after the conclusion of the conference and completion of the post-conference survey

1.3 Acknowledgments

- To the National Association of Psychometrists (NAP) Conference Planning Committee and all organizers of this conference
- To my colleagues, Amelia VanderLeest and Blair Shaw
- To the other presenters, before and after me
- To you, my audience

1.4 Learning Objectives

Learning objectives below have been approved by NAP and the Board of Certified Psychometrists (BCP). These objectives will be met with an emphasis on relevancy to the discipline of psychometry.

- Learn the general, basic differences between the frequentist and Bayesian frameworks for statistics
- Review the historical and modern prevalent use of frequentist statistics in neuropsychology and cognitive testing
- Appreciate the various developing applications of Bayesian statistics to neuropsychological evaluation - Consider the practical ramifications in test selection and scoring that come with a Bayesian perspective, along with the possible future directions for this approach

- Understand the limitations and ramifications of implementing these methods in a clinic

1.5 My Pitch

- Empirical and evidence-driven nurse analogy
  - We owe it to our patients and clinics to be well-versed in empirical knowledge, even if not used daily
- Improving our stature, variety, and skills
  - With an ever-evolving landscape involving digital testing, AI, and new technology, we must be mindful of how we can train ourselves up in new methods, many of which have roots in Bayesian statistics
- Becoming skilled in statistics may offer an important avenue to future research and educational opportunities!

2 Frequentist vs. Bayesian

3 Prevalent Use of Frequentist Statistics

4 Neuropsychological Applications of the Bayesian Method

5 Limitations of Bayesian Methods in the Clinic and Research

6 Conclusion

6.1 Further Reading & Resources

- Books
  - For a friendly and more basic introduction to the Bayesian Method → *A Student's Guide to Bayesian Statistics* (Lambert, 2018)
  - For broad coverage of statistics topics leading up to Bayesian Methods → *Statistical Rethinking, 2nd Edition* (McElreath, 2020)
  - For more focused applications of Bayesian Methods to psychology → *Bayesian Statistics for the Social Sciences, 2nd Edition* (Kaplan, 2024)
- Review the References of this presentation for scientific articles that show these methods in action!

6.2 Take-home Message

- Bayesian statistics differ greatly from frequentist statistics in the framing and questioning of probabilities
- While frequentist statistics do remain prominent, advanced methodologists are seeing more innovating ways to apply Bayesian methods to success in neuropsychological research
- Because of their unique position, Bayes-inspired methods offer a meaningful way to pursue new advances in research and clinical thinking
- While innovative, there are still many challenges in practically applying these ideas to the daily work of psychometrists

6.3 Contact Information

Thank you for your attention today - I welcome you to stay in touch with me to talk about this topic and other exciting neuropsychology!

- Email: Quinton.Quagliano@trinity-health.org
- Find me on LinkedIn!

Any Questions?

7 Technical Information

7.1 System and Program Information

Replicability of this file is dependent on having all necessary system dependencies. These include, but are not limited to, a TeX distribution, Pandoc, R, all listed R packages and their dependencies, and many more. Please ensure that your system is capable of knitting other Quarto documents and running R code.

Alternatively, the build environment can be replicated using the included nix flake file, which can replicate the exact TeX, Quarto, and R package versions used in this analysis. Please see the section on Reproducability Information.

This work has been version controlled with git, to improve ability to review history of files. All source code, files needed to reproduce, and .git files will be included with this document for transparency. The source code used to generate this document, and it's accompanying presentation, can be found on Github.

See Table 1 for system, R, and package information regarding the system that this document was originally rendered on.

Table 1: Technical Information

(a) System and R Information

	Specifications
platform	x86_64-pc-linux-gnu
arch	x86_64
os	linux-gnu
system	x86_64, linux-gnu
status	
major	4
minor	5.3
year	2026
month	03
day	11
svn.rev	89597
language	R
version.string	R version 4.5.3 (2026-03-11)
nickname	Reassured Reassurer

(b) Packages

Package	Version
magrittr	2.0.4
sessioninfo	1.2.3

7.2 Reproducability Information

Steps have been taken to make this document reproducible and able to be audited. Prior to attempting the steps in reproducing this work and generated PDF/presentation, please acquire/install any necessary tools and packages, as listed in Section 7.1. Because of limitations in testing time, it cannot be guaranteed that this work is entirely reproducible - please contact the author using information from the Contact Information section to get in touch if anything seems off.

8 References

Kaplan, D. (2024). *Bayesian statistics for the social sciences* (Second edition). The Guilford Press.

Lambert, B. (2018). *A student's guide to Bayesian statistics*. SAGE.

McElreath, R. (2020). *Statistical rethinking: A Bayesian course with examples in R and Stan* (Second edition). CRC Press.